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Sem : 7

Div : A

Rollno : 19

Subject : Application Development Using Full Stack-701

Assignment-1

Practical_Assignmnet : 1

Question 1 :

Develop nodejs application with following requirements:

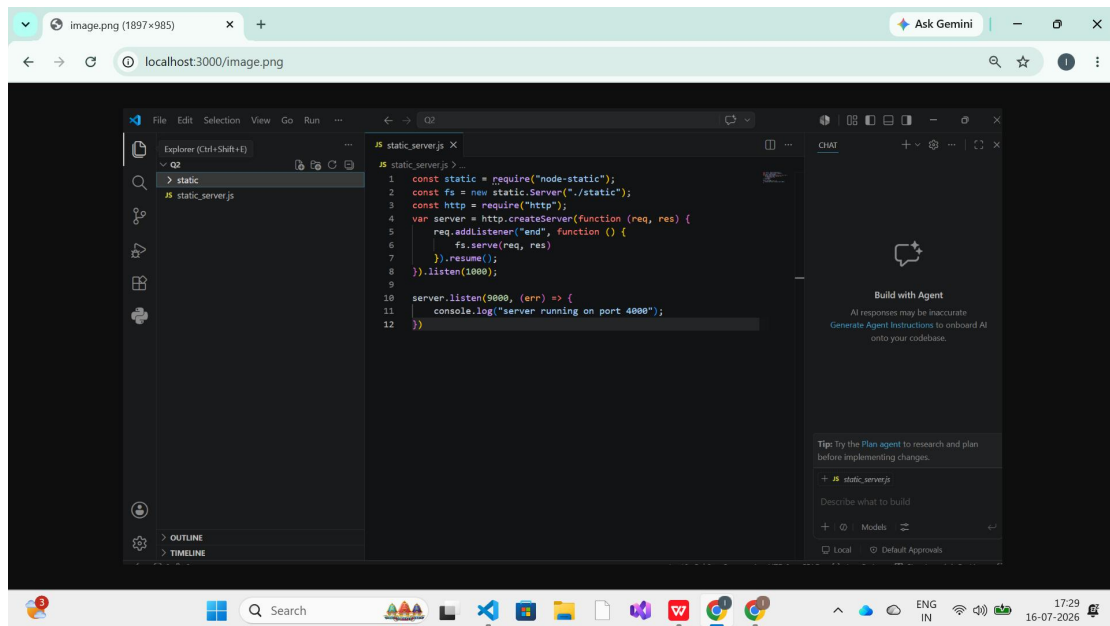
- Develop a route "/gethello" with GET method. It displays "Hello NodeJS!!" as response.
- Make an HTML page and display.
- Call "/gethello" route from HTML page using AJAX call. (Any frontend AJAX call API can be used.)



Hello Nodejs !!

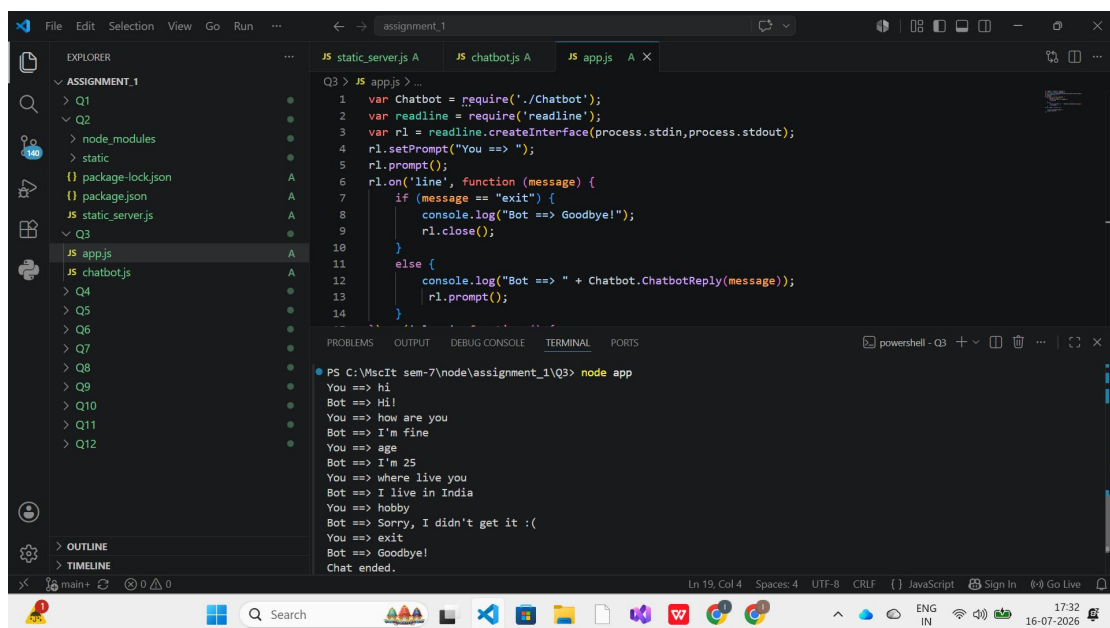
Question 2 :

Develop a web server which serves static resources.



Question 3 :

Develop a module for domain specific chatbot and use it in a command line application.



Question 4 :

Write a program to create a compressed zip file for a folder.

The screenshot shows a VS Code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'assignment_1' with a folder 'sample' containing 'image.png', 'image.png.gz', 'sample.txt', and 'sample.txt.gz'. The code editor shows a file named 'compress_zip.js' with the following code:

```
1 //Write a program to create a compressed zip file for a folder.
2 const fs = require("fs");
3 const zlib = require("zlib");
4 const path = require("path");
5 const files = fs.readdirSync("./sample");
6
7 files.forEach((file) => {
8     const inputPath = path.join("./sample", file);
9
10    if (fs.statSync(inputPath).isFile()) {
11        const input = fs.createReadStream(inputPath);
12
13        const output = fs.createWriteStream(
14            path.join("./sample", file + ".gz")
15        );
16
17        input
```

The terminal at the bottom shows the following commands and output:

```
PS C:\VscIt sem-7\node\assignment_1> cd Q4
PS C:\VscIt sem-7\node\assignment_1\Q4> node compress_zip
sample.txt compressed successfully
image.png compressed successfully
PS C:\VscIt sem-7\node\assignment_1\Q4>
```

Question 5 :

Write a program to extract a zip file.

The screenshot shows a VS Code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'assignment_1' with a folder 'sample' containing 'extracted_sample.txt', 'sample.txt', and 'sample.txt.gz'. The code editor shows a file named 'file_extract.js' with the following code:

```
1 //Write a program to extract a zip file.
2
3 const fs=require("fs");
4 const zlib=require("zlib");
5 const input=fs.createReadStream("sample.txt.gz");
6 const output=fs.createWriteStream("extracted_sample.txt");
7
8 input.pipe(zlib.createGunzip())
9     .pipe(output)
10    .on("finish",()=>{
11        console.log("file extracted successfully");
12    })
```

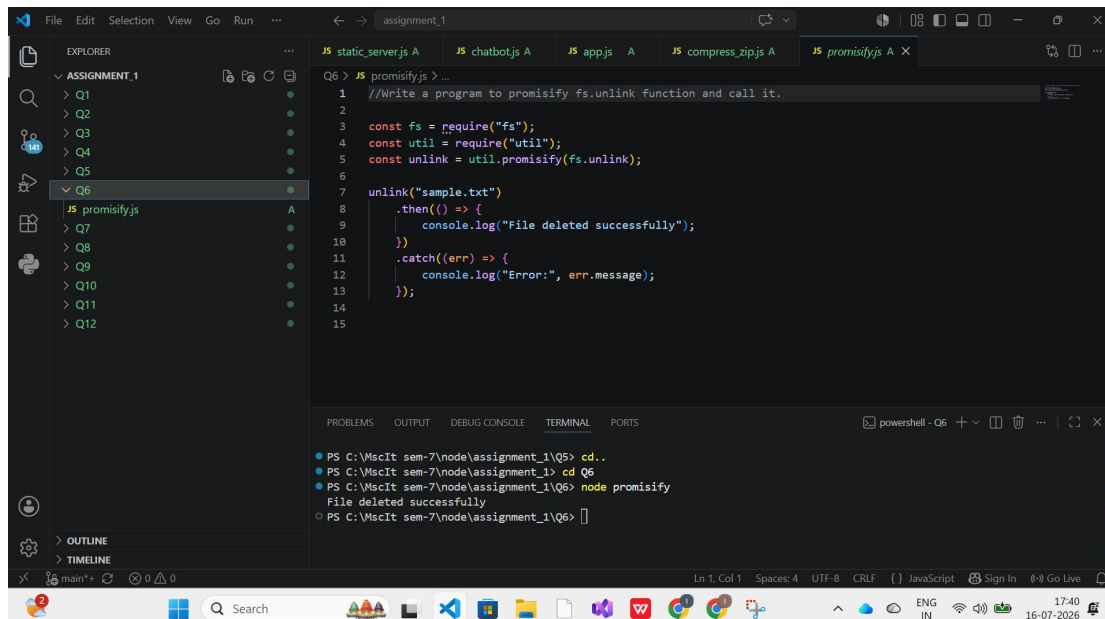
The terminal at the bottom shows the following commands and output:

```
PS C:\VscIt sem-7\node\assignment_1\Q4> node file_extract.js
code: 'MODULE_NOT_FOUND',
requireStack: []
}

Node.js v24.17.0
PS C:\VscIt sem-7\node\assignment_1\Q4> cd..
PS C:\VscIt sem-7\node\assignment_1> cd Q5
PS C:\VscIt sem-7\node\assignment_1\Q5> node file_extract
file extracted successfully
PS C:\VscIt sem-7\node\assignment_1\Q5>
```

Question 6 :

Write a program to promisify fs.unlink function and call it.

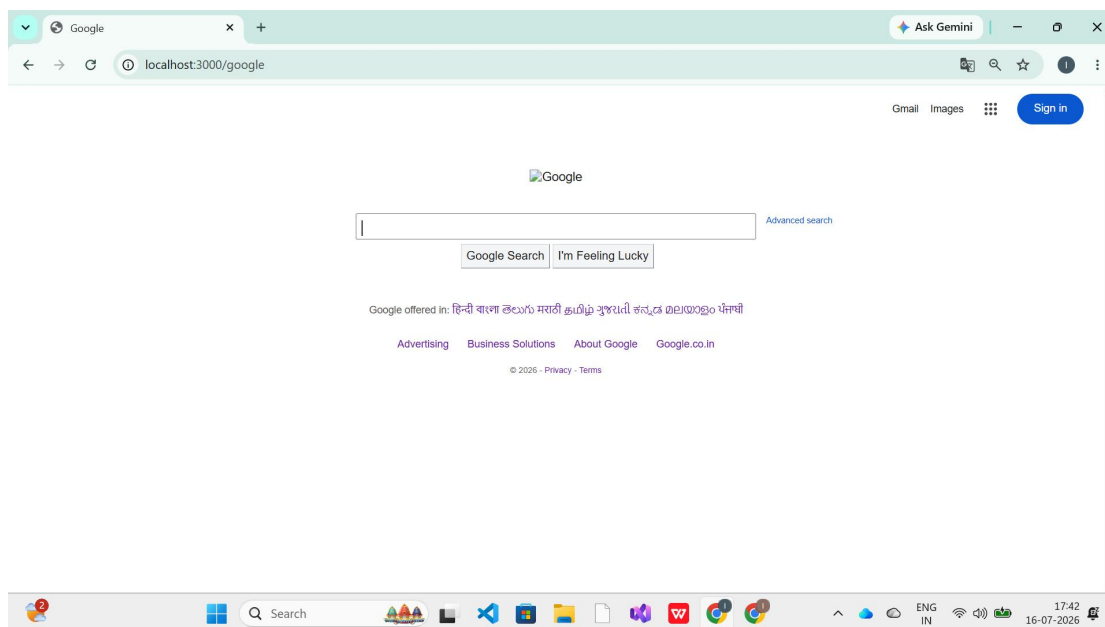


```
Q6 > JS promisifyjs > ...
1 //Write a program to promisify fs.unlink function and call it.
2
3 const fs = require("fs");
4 const util = require("util");
5 const unlink = util.promisify(fs.unlink);
6
7 unlink("sample.txt")
8   .then(() => {
9     console.log("File deleted successfully");
10  })
11  .catch((err) => {
12    console.log("Error:", err.message);
13  });
14
15
```

```
PS C:\WscIt sem-7\node\assignment_1\Q5> cd..
PS C:\WscIt sem-7\node\assignment_1> cd Q6
PS C:\WscIt sem-7\node\assignment_1\Q6> node promisify
File deleted successfully
PS C:\WscIt sem-7\node\assignment_1\Q6>
```

Question 7 :

Develop a node web server with a route /google which can fetch data of google page using fetch() using async-await model and show as a response.



Question 8 :

Set a server script, a test script and 3 user defined scripts in package.json file in your nodejs

The screenshot shows the Visual Studio Code editor with the 'assignment_1' workspace. The Explorer sidebar on the left shows a folder named 'ASSIGNMENT_1' containing subfolders Q1 through Q12, and files build.js, clean.js, dev.js, package.json, server.js, and test.js. The package.json file is open in the editor, showing the following content:

```
1 {
2   "name": "q8",
3   "version": "1.0.0",
4   "description": "",
5   "main": "index.js",
6   "scripts": {
7     "test": "node test.js",
8     "start": "node server.js",
9     "build": "node build.js",
10    "clean": "node clean.js",
11    "dev": "node dev.js"
12  },
13  "author": ""
14}
```

The terminal at the bottom shows the execution of the following commands:

```
PS C:\VscIt sem-7\node\assignment_1\Q8> npm run dev
> q8@1.0.0 dev
> node dev.js

Development mode started...

PS C:\VscIt sem-7\node\assignment_1\Q8> npm run build
> q8@1.0.0 build
> node build.js

Building application...
```

Question 9 :

A program which calls useful functions in fs module.

The screenshot shows the Visual Studio Code editor with the 'assignment_1' workspace. The Explorer sidebar on the left shows the 'file.js' file selected. The file.js file is open in the editor, showing the following content:

```
1 Hello Node.js
2 I am appending data to the file
3
```

The terminal at the bottom shows the execution of the following commands:

```
PS C:\VscIt sem-7\node\assignment_1\Q9> node file
....File created successfully....
....Data appended successfully....

File reading successfully
Hello Node.js
I am appending data to the file

....file exists....
Size: 46 bytes
File: true
Directory: false

....File renamed successfully....

....File copied successfully....

....File Deleted successfully....
```

Question 10 :

A program which uses global objects in nodejs. Also read command line arguments and print it.

The screenshot shows a Visual Studio Code editor window with a file named `global_objects.js` open. The file contains the following code:

```
1 // __filename
2 console.log("\nCurrent file path :",__filename);
3
4 // __dirname
5 console.log("\nCurrent directory :",__dirname);
6
7 // process object
8 console.log("\nVersion : ",process.version);
9 console.log("\nOperating system : ",process.platform);
10
11 // Command Line Arguments
```

The terminal output shows the execution of the file:

```
PS C:\MscIt sem-7\node\assignment_1\Q10> node file
....File Deleted successfully.
PS C:\MscIt sem-7\node\assignment_1\Q9> cd ..
PS C:\MscIt sem-7\node\assignment_1> cd Q10
PS C:\MscIt sem-7\node\assignment_1\Q10> node global_object

Current file path : C:\MscIt sem-7\node\assignment_1\Q10\global_object.js
Current directory : C:\MscIt sem-7\node\assignment_1\Q10
Version : v24.17.0
Operating system : win32
Arguments :
No arguments provided.
PS C:\MscIt sem-7\node\assignment_1\Q10>
```

Question 11 :

Develop a useful package and publish it on npmjs.com. Write a program to use it.

Question 12 :

. Write a node program having at least 3 dependent tasks and call using chained promises.

The screenshot shows a Visual Studio Code editor window with a file named `promise_isha_khunt_7` open. The file contains the following code:

```
1 // __filename
2 console.log("\nCurrent file path :",__filename);
3
4 // __dirname
5 console.log("\nCurrent directory :",__dirname);
6
7 // process object
8 console.log("\nVersion : ",process.version);
9 console.log("\nOperating system : ",process.platform);
10
11 // Command Line Arguments
```

The terminal output shows the execution of the file:

```
PS C:\MscIt sem-7\node\assignment_1> cd Q12
PS C:\MscIt sem-7\node\assignment_1\Q12> node promise_isha_khunt_7
Task 1: Student Registered
...Registration Completed...
Task 2: Fees Paid
...Fee Payment Completed....
Task 3: Fee receipt generate
....Fees Status updated....
PS C:\MscIt sem-7\node\assignment_1\Q12>
```

